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MSSE 601 - 26F8W1
Week 4 on UI Design

Portfolio Decision Record

Context

During Week 4, the team was responsible for producing user interface designs for the Golden Bike Shop application. I owned the user interface design deliverable and was responsible for creating the initial screen designs, user journeys, team decision record, and AI use record. As part of the assignment requirements, the team needed to evaluate AI-generated alternatives and determine whether those designs should be incorporated into the final submission.

The system supports several different employee roles, including retail managers, rental employees, and service staff members. Users needed interfaces that would allow them to complete inventory lookup, rental management, and service scheduling tasks efficiently. At the same time, the project requirements emphasized a clear and consistent user interface that trained employees could use without technical assistance. The situation that forced a decision was when we asked A.I. to generate some sample screens after feeding in our screen mockups. It did come back a lot of features that looked convoluted and adding a lot of complexity that this application is not set up for, however, the sidebar menu was visually appealing and would make it easier for a store manager who has more access to many of the features to easily navigate between multiple sections. The challenge was deciding whether to adopt the more visually complex AI-generated screens or continue with the simpler layouts that had already been drafted from the project requirements and user journeys.

Options

Option 1: Adopt the AI-generated designs

The AI-generated alternatives included additional visual elements such as large product images, bike icons, split-screen layouts, dashboard views, map-based bike tracking displays, and more detailed information presented directly on the main screens. Some alternatives also introduced guided workflows and additional visual navigation elements.

Option 2: Keep the original designs and selectively adopt useful AI ideas

This option retained the screen layouts already drafted by the team while evaluating individual AI suggestions on their own merits. Useful additions could be incorporated if they improved usability or supported system requirements, while unnecessary complexity would be rejected.

Decision

I recommended that the team keep the original screen designs and selectively adopt only the AI-generated features that improved usability while remaining aligned with the project requirements. The team accepted this approach. We incorporated features such as inventory dates, filtering options, navigation menus, and clearer out-of-stock indicators. At the same time, we rejected large product icons, map-based dashboards, split-screen layouts, and other features that added complexity without supporting stated stakeholder needs.

Reasoning

The primary factor that drove this decision was usability. The requirements specified that the system should provide a clear and consistent interface that allows employees to complete their primary tasks without assistance. NFR-05 specifically emphasizes usability and consistency across the application. Many of the AI-generated alternatives placed additional information directly on the screen or relied heavily on visual elements. While visually appealing, these designs increased cognitive load and could distract users from the actions they needed to perform. For example, inventory lookup primarily required users to locate inventory quantities, locations, and stock status. Large images of products would occupy significant screen space while providing little value for the task being performed.

The same reasoning applied to the rental and service management screens. The map-based rental dashboard presented information that was not required by any stated requirement, while the service-screen alternatives focused on details outside the scheduling workflow that the assignment emphasized. Since the project requirements focused on renting available bikes, viewing telemetry, and managing service bay schedules, the simpler interfaces better supported the actual user tasks. However, the AI alternatives were still useful because they highlighted improvements we had overlooked. Features such as filter controls, navigation menus, inventory dates, and clearer status indicators improved usability without sacrificing simplicity. These additions strengthened the design while keeping the user experience focused on the requirements.

Consequences

The decision resulted in interfaces that were easier to understand and more closely aligned with the project requirements. The screens remained focused on inventory management, rentals, and service scheduling rather than visual presentation. This also reduced implementation complexity because the system would not require storing and managing large numbers of product or bicycle images. The main tradeoff was that we gave up potentially richer visual experiences. Features such as map-based tracking or image-heavy inventory displays might make some information easier to recognize at a glance. We also gave up opportunities to explore more innovative interface concepts that could become useful in a larger commercial system.

With what I know now, I would revisit this decision if future requirements introduced GPS tracking, visual bike identification, or other features where maps and images directly support user tasks. Under the current requirements, however, the simpler interface was the better choice because it prioritized clarity, usability, and alignment with stakeholder needs.

Evidence

Week Owned: Week 4 - User Interface Design Project Component

Deliverable: Kickstarters_MSSE601_UIDesign

Where the Decision Appears: Team Decision Record, "Decision 4 - Incorporating AI-Generated Alternative Designs" and the AI Use Record.

AI Contribution: Copilot generated multiple alternative interface designs and suggested additional usability features. I accepted inventory dates, filters, navigation menus, and status indicators, while rejecting map-based dashboards, large icons, and other features that increased complexity without supporting project requirements.